Momentum Call Strategy

Systematic Equity Options — Buy Call Approach

Prepared June 21, 2026

Backtest Period: September 2020 – February 2026 (~5.5 years)

Dynamic Universe — Top 30 Stocks | *S&P 500*

1. Strategy Overview

The Momentum Call Strategy is a systematic, rules-based approach that uses call options to capture upward price momentum in large-cap U.S. equities. The strategy monitors a dynamic universe of 30 stocks — selected each period from the S&P 500 for their strong options activity and liquidity — and enters call positions only on names showing a confirmed positive price trend. Capital not deployed in options is kept in short-term instruments, generating a steady income base independent of market direction.

How it works:

• **Momentum filter:** each period, a stock qualifies for a call position only if its previous monthly candle closed higher than it opened — a simple, objective confirmation of upward trend.

• **Call option entry:** an at-the-money call is purchased at the prior month's closing price as the strike. The option captures the next move up with a defined, limited cost.

• **Average option spend:** ~1.52% of notional per period across the portfolio. This figure is naturally low because not all 30 stocks pass the momentum filter simultaneously — on average 15–20 names are active each period, so only a fraction of capital is committed to options at any one time.

• **Cash reserve:** the substantial portion of capital not deployed in options is held in short-term instruments, earning prevailing market interest rates (0.08% p.a. during 2020–21 near-zero rates; 4.3–5.1% p.a. post-2022).

• **Equal weighting:** all active positions carry the same weight, avoiding concentration risk in any single name.

• **Rebalance cadence:** the entire portfolio is reviewed and reset every ~3–4 weeks, aligned with monthly option expiry cycles.

Structural edge: because call options offer unlimited upside with strictly capped downside, the strategy benefits from positive return asymmetry — winning periods tend to be significantly larger than losing ones. The income from the cash reserve provides a meaningful buffer, reducing the net cost of holding options and dampening portfolio drawdowns.

2. Strategy Parameters

Parameter	Value
Investment universe	S&P 500 — 50 large-cap stocks across 10 sectors
Active portfolio size	Top 30 most liquid, high-activity names per period
Entry condition	Stock's prior monthly candle closes UP (close > open)
Instrument	At-the-money call option, strike = prior month close
Holding period	~3–4 weeks (one monthly expiry cycle)
Avg option cost	~1.52% of notional (reflects partial signal activation — typically 15–20 of 30 stocks qualify each period)
Cash reserve	Balance of capital held in short-term instruments
Cash yield (2020–21)	~0.08% p.a. (low-rate environment)
Cash yield (2022)	~2.00% p.a. (rate normalisation)
Cash yield (2023–26)	~4.30–5.10% p.a. (current rate environment)
Position sizing	Equal weight across all qualifying positions
Rebalance frequency	Monthly — full portfolio reset each expiry
Backtest period	September 2020 – February 2026 (72 periods, ~5.5 years)

3. Performance & Risk Statistics

Two capital-deployment variants are shown side by side. **Unlevered** deploys 1x notional in the options book; idle, undeployed capital earns a plain money-market rate (no leverage on the cash leg). **4x Leveraged** applies 4x notional to the same option signals, scaling both the option P&L and the option cost by 4x, with any capital still idle after the 4x option allocation earning money-market. Over **74 trading periods** (2020-10-05 to 2026-02-13), Unlevered returned **+47.4%** total (CAGR +7.1%, max drawdown -3.3%), while 4x Leveraged returned **+131.9%** total (CAGR +15.9%, max drawdown -14.0%). Leverage amplifies both the gains and the drawdowns roughly proportionally — it is not a free source of additional return.

Metric	Unlevered (1x)	4x Leveraged
Total Return	**+47.38%**	**+131.90%**
CAGR	**+7.05% p.a.**	**+15.92% p.a.**
Max Drawdown	**-3.26%**	**-13.97%**
Sharpe Ratio	**1.68**	**1.18**
Sortino Ratio	**2.13**	**1.43**
Win Rate	**52/74 (70%)**	**48/74 (65%)**
Avg return / period	**+0.640%**	**+1.782%**
Volatility / period	**1.378%**	**5.452%**
Best period	**2024-10-28 +4.23%**	**2024-10-28 +15.74%**
Worst period	**2022-08-08 -2.78%**	**2022-08-08 -11.59%**
Avg option cost	**1.94% of notional**	**1.94% of notional**

4. Growth of Capital

The charts below show how $100 of initial capital would have grown over the backtest period under each variant, expressed as cumulative percentage return. The lower panel of each tracks the maximum decline from any prior peak. Leverage scales both the upside and the drawdowns — it does not change the underlying win rate or signal quality, only the size of each swing.

Momentum Call Strategy | Signal-Filtered (Cap 30) | S&P 500 Universe | Unlevered — Idle Cash Earns Money-Market

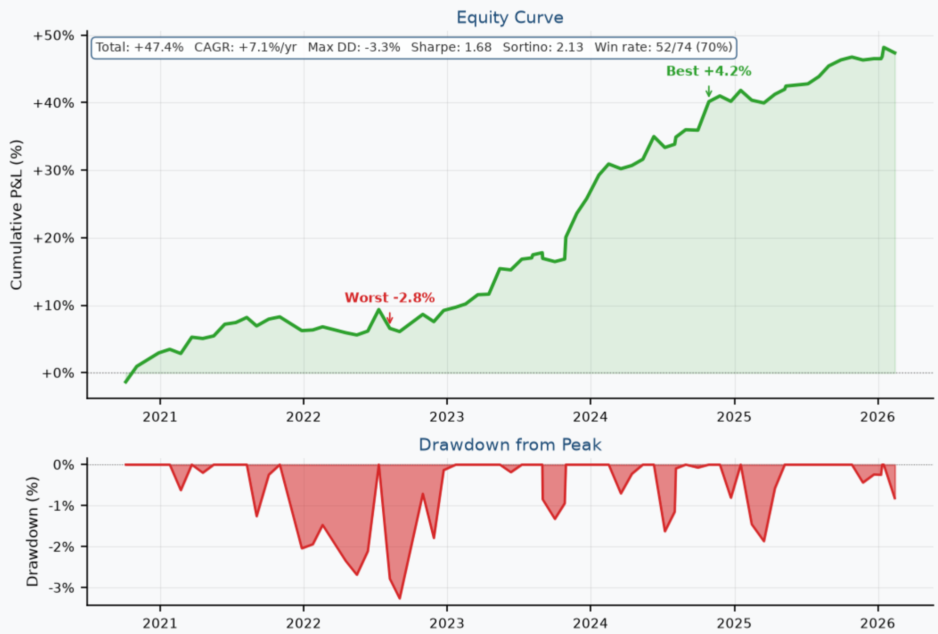

Fig. 1a — Unlevered: cumulative return (%) and drawdown, Sep 2020 – Feb 2026

Momentum Call Strategy | Signal-Filtered (Cap 30) | S&P 500 Universe | 4x Leveraged — Idle Cash Earns Money-Market

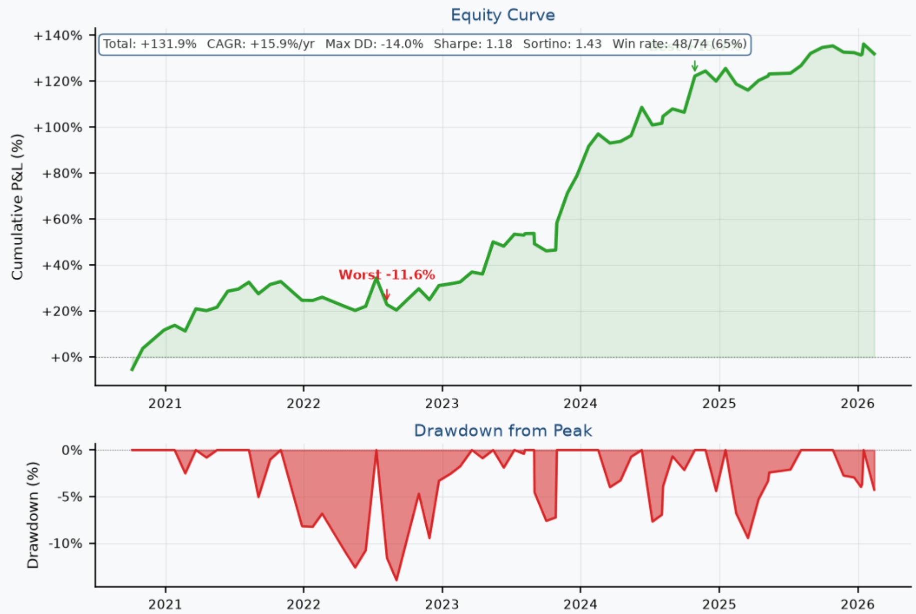

Fig. 1b — 4x Leveraged: cumulative return (%) and drawdown, Sep 2020 – Feb 2026

5. Monthly Returns at a Glance

The heatmaps below present strategy returns for each calendar month across the full backtest, for both variants. Each cell shows the return for that period. **Green** cells represent positive months, **red** cells represent negative months, and yellow indicates near-zero. The Annual column shows the full-year total.

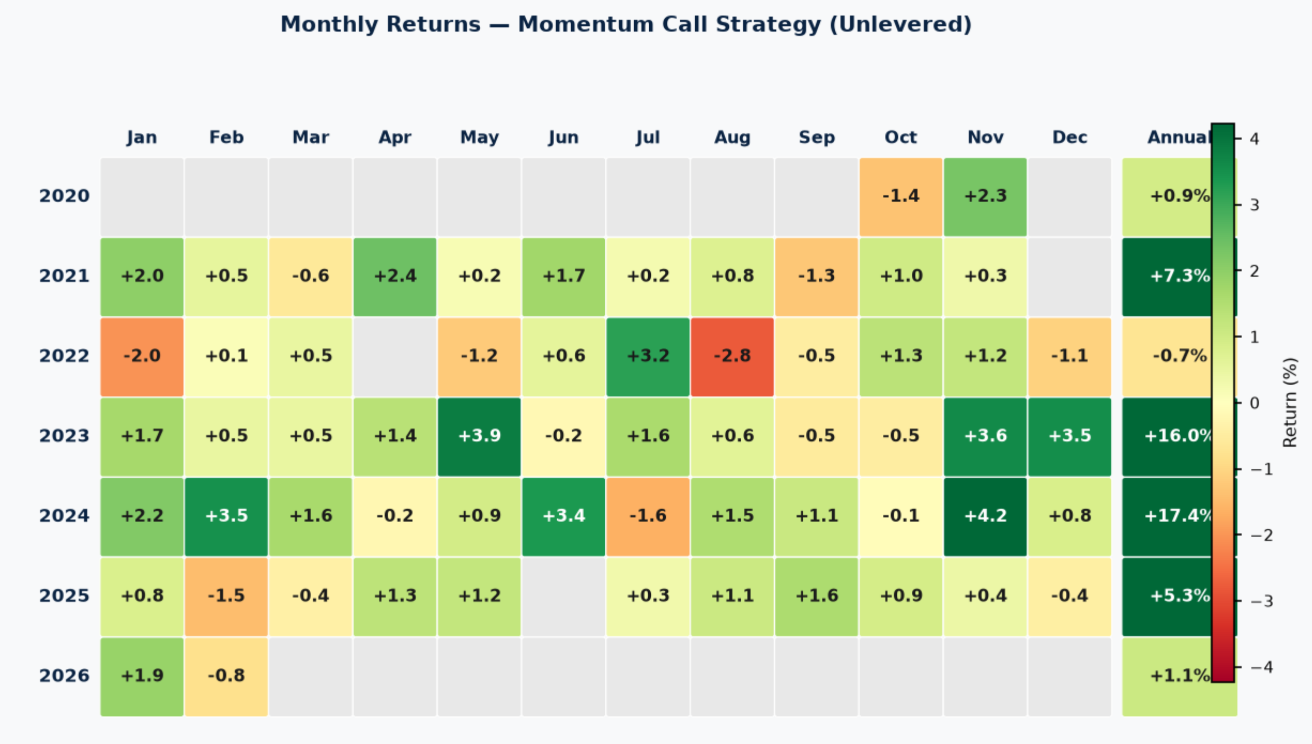

Fig. 2a — Unlevered monthly return heatmap (%), Sep 2020 – Feb 2026

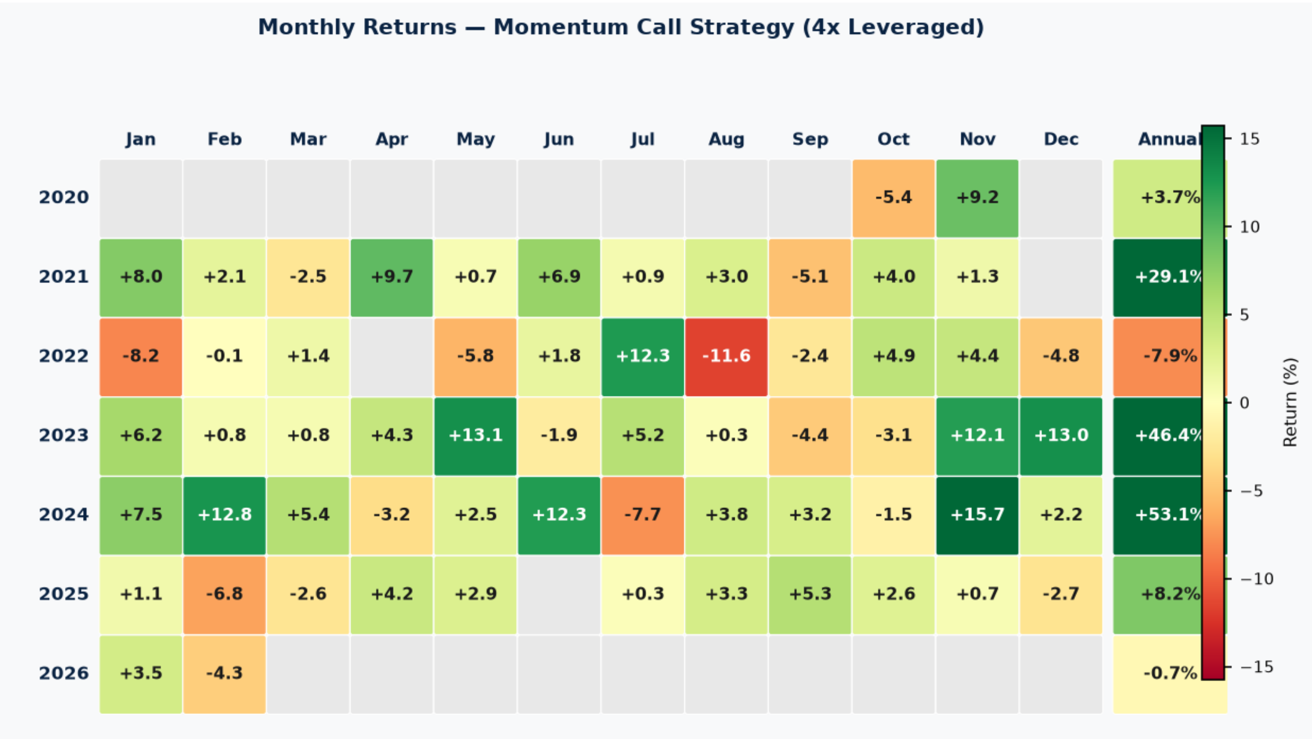

Fig. 2b — 4x Leveraged monthly return heatmap (%), Sep 2020 – Feb 2026

6. Monthly Returns — Detail Table (Unlevered)

Full numeric breakdown of unlevered returns by month and year — 1x notional in the options book, idle capital earning plain money-market, no leverage. Colour coding matches the heatmap: green = positive, red = negative, yellow ≈ zero.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual
2020	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	-1.4	+2.3	+nan	**+0.9**
2021	+2.0	+0.5	-0.6	+2.4	+0.2	+1.7	+0.2	+0.8	-1.3	+1.0	+0.3	+nan	**+7.3**
2022	-2.0	+0.1	+0.5	+nan	-1.2	+0.6	+3.2	-2.8	-0.5	+1.3	+1.2	-1.1	**-0.7**
2023	+1.7	+0.5	+0.5	+1.4	+3.9	-0.2	+1.6	+0.6	-0.5	-0.5	+3.6	+3.5	**+16.0**
2024	+2.2	+3.5	+1.6	-0.2	+0.9	+3.4	-1.6	+1.5	+1.1	-0.1	+4.2	+0.8	**+17.4**
2025	+0.8	-1.5	-0.4	+1.3	+1.2	+nan	+0.3	+1.1	+1.6	+0.9	+0.4	-0.4	**+5.3**
2026	+1.9	-0.8	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	+nan	**+1.1**

7. Recent History — Last 6 Periods (Unlevered)

Detail view of the most recent monthly cycles under the unlevered (1x) variant, showing how many tickers actually signaled BUY CALL that period, how many were active (capped at 30, prioritized by premium %), the average option premium paid, the option P&L, the money-market contribution from idle cash, and the resulting period and cumulative returns. This is the same mechanic that drives every period in the backtest — shown here at full granularity as a worked example. (A 4x leveraged variant exists — see Section 3 — and scales the option P&L and option cost columns by 4x; idle-cash credit shrinks accordingly.)

Period	Signals	Active	Avg Premium	Option P&L	MM Contrib	Total P&L	Cumulative
2025-11-24	11	11/30	1.04%	-0.76%	+0.33%	**-0.44%**	+46.3%
2025-12-22	27	27/30	2.22%	-0.13%	+0.32%	**+0.20%**	+46.5%
2026-01-09	1	1/30	0.34%	-0.34%	+0.33%	**-0.01%**	+46.5%
2026-01-12	1	1/30	0.20%	-0.01%	+0.33%	**+0.32%**	+46.8%
2026-01-16	17	17/30	1.79%	+1.04%	+0.32%	**+1.37%**	+48.2%
2026-02-13	15	15/30	1.54%	-1.15%	+0.33%	**-0.82%**	+47.4%

Most recent period — tickers held:

AAPL, ABBV, CMCSA, COST, CVX, HD, JNJ, KO, LLY, META, MRK, PEP, PG, WMT, XOM

8. Investment Case

• **Unlevered: +47.4% total return over 5.5 years — CAGR +7.1% p.a.**, with a peak drawdown of 3.3%. This is the realistic base case: only tickers that actually signal BUY CALL are traded (capped at 30 active slots, prioritized by option premium %), idle capital earns plain money-market, no leverage on either leg.

• **4x Leveraged: +131.9% total return — CAGR +15.9% p.a.**, with a peak drawdown of 14.0%. Leverage scales both the return and the drawdown versus the unlevered case — it amplifies the existing edge, it does not create a new one.

• **Call options provide built-in downside protection.** Unlike direct equity exposure, the maximum loss on any single position is limited to the option cost (scaled by leverage, if used). This structural feature caps the damage in adverse periods while preserving full participation in strong moves.

• **The cash reserve is a real contributor to returns, not leveraged.** Money-market income only accrues on capital that is genuinely idle after the options allocation — never on borrowed or notional exposure — and uses period-appropriate short-term rates (0.08% in 2020–21 ZIRP, rising to ~5% in 2023–24).

• **Sharpe / Sortino — Unlevered 1.68 / 2.13, 4x Leveraged 1.18 / 1.43.** The unlevered variant has the better risk-adjusted profile on both measures; leverage increases the absolute return but at a worse Sharpe/Sortino ratio, as expected when scaling a fixed signal with a fixed-cost cash leg.

• **Win rate: 70% unlevered (52/74 periods), 65% with 4x leverage (48/74)** — the same underlying trade decisions; leverage changes the magnitude of wins and losses, not which periods are profitable.

• **By calendar year (unlevered / 4x leveraged): 2022 −0.7% / −7.9%, 2023 +16.0% / +46.4%, 2024 +17.4% / +53.1%.** 2023–24 broad market momentum drove the bulk of cumulative return; 2022's rate-shock year was the one calendar-year loss in this backtest, amplified under 4x leverage. (Each ~4-week period is attributed to the calendar month containing its midpoint — see Section 6 for the full monthly breakdown, which sums exactly to these yearly figures.)

• **Fully systematic and low-maintenance.** Signals are generated once per monthly cycle with no intraday monitoring required. The approach is transparent, rule-based, and straightforward to audit.